

. ECE 1387: CAD For Digital Circuit Synthesis and Layout

- J. Anderson
- Fall 2025

. Reading List

- Lecture 1: Introduction and Overview
 - Background literature:
 - Gordon E. Moore, "Cramming More Components onto Integrated Circuits," *Electronics*, pp. 114–117, April 19, 1965. ([PDF](#))
- Lecture 2: Detailed Routing (applied to FPGAs)
 - Background Literature:
 - C.Y. Lee, "An algorithm for path connections and its applications," *IRE Transactions on Electronic Computers*, vol. 10, pp. 346-365, Sept. 1961. ([PDF](#)) (Classic paper on maze routing)
 - See Section III-D of: M.A. Breuer, M. Sarrafzadeh, F. Somenzi, "Fundamental CAD algorithms," *IEEE Transactions on CAD*, Vol. 19, No. 12, pp. 1449-1475, December 2000 ([PDF](#))
 - J.S. Rose, S.D. Brown, "Flexibility of interconnection structures for field-programmable gate arrays", *IEEE Journal of Solid State Circuits*, Vol. 26 No. 3, pp. 277-282, March 1991. (reference only; classic paper that may help with assignment #1) ([PDF](#))
 - S.D. Brown, J.S. Rose, Z.G. Vranesic, "A detailed router for field-programmable gate arrays," *IEEE Transactions on CAD*, Vol. 11, No. 5, pp. 620-628, May 1992. (reference only: may help with assignment #1) ([PDF](#))
- Lecture 3: Timing-Driven Routing
 - Background literature:
 - Ravi Nair, "A simple yet effective technique for global wiring," *IEEE Transactions on CAD*, Vol. 6, No. 2, pp. 165-172, 1987. ([PDF](#))
 - Larry McMurchie, Carl Ebeling. "PathFinder: A negotiation-based performance-driven router for FPGAs," *ACM International Symposium on Field Programmable Gate Arrays*, pp. 111-117, 1995. ([PDF](#))
 - Jordan S. Swartz, Vaughn Betz, Jonathan Rose: A Fast Routability-Driven Router for FPGAs. *FPGA 1998*: 140-149 ([PDF](#)).
 - R. Fung, V. Betz, and W. Chow, "Slack Allocation and Routing to Improve FPGA Timing While Repairing Short-Path Violations," *IEEE Trans. on Computer-Aided Design of Circuits and Systems*, April 2008, pp. 686 - 697. ([PDF](#))
 - M. Gort, and J. Anderson, "Combined Architecture/Algorithm Approach to Fast FPGA Routing," *IEEE Trans. on VLSI*, Vol. 21, No. 6, pp. 1067-1079, June 2013 ([PDF](#))
 - M. Stojilovic, "Parallel FPGA Routing: Survey and Challenges," *Int'l Conf. on Field-Programmable Logic and Applications*, 2017 ([PDF](#))
- Lecture 4-5: Routing/Placement
 - Background literature:
 - M. Kim, D. Lee, I. Markov, "SimPL: An algorithm for placing VLSI circuits," *Communications of the ACM*, Vol 56, no 6., June 2013 ([PDF](#)).
 - Paper on the Bound2Bound net model ([PDF](#))
 - M. Gort, J. Anderson, "Analytical placement for heterogeneous FPGAs," *IEEE FPL*, 2012, pp. 143-150 ([PDF](#)).
 - Nima Karimpour Darav, Andrew A. Kennings, Kristofer Vorwerk, Arun Kundu: Multi-Commodity Flow-Based Spreading in a Commercial Analytic Placer. *FPGA 2019*: 122-131. ([PDF](#))
- Lecture 6-7: Simulated Annealing-Based Placement
 - Background literature:
 - S. Kirkpatrick, C. Gellat, M. Vecchi, "Optimization by simulated annealing," *Science*, 220:671-680, 1983. ([PDF](#))
 - C. Sechen, A. Sangiovanni-Vincentelli, "The TimberWolf placement and routing package," *IEEE Journal of Solid State Circuits*, Vol. 20, No. 2, pp. 432-439, 1985. ([PDF](#))
 - V. Betz, J. Rose, "VPR: A new packing, placement and routing tool for FPGA research," *International Workshop on Field-Programmable Logic and Applications*, London, UK, 1997, pp. 213 -

222. ([PDF](#))

- A. Ludwin and V. Betz, "Efficient and Deterministic Parallel Placement for FPGAs," ACM Trans. on Design Automation of Electronic Systems, Vol. 16, No. 3, June 2011, pp. 22:1 - 22:23. ([PDF](#))
- G. Zhou, M. Stojilovic, J. Anderson, "GRAMM: Fast CGRA Application Mapping Based on A Heuristic for Finding Graph Minors," FPL 2023 ([PDF](#))

- Lecture 8-9: Partitioning (Branch and bound, FM + multi-level)

- Background Literature:

- C. M. Fiduccia, R. M. Mattheyses. "A linear-time heuristic for improving network partitions," *ACM/IEEE Design Automation Conference*, pp 174-181, 1982. ([PDF](#))
- G. Karypis, R. Aggarwal, V. Kumar, S. Shekhar, "Multilevel hypergraph partitioning: Applications in the VLSI domain," *IEEE Transactions on VLSI Systems*, 07(01):69-79, March 1999. ([PDF](#))
- Lars W. Hagen, Dennis J.-H. Huang, Andrew B. Kahng: On implementation choices for iterative improvement partitioning algorithms. *IEEE Trans. on CAD of Integrated Circuits and Systems* 16(10): 1199-1 205 (1997) ([PDF](#))

- Lecture 10: Technology Mapping (Dynamic Programming)

- Background Literature:

- A. Mishchenko, S. Chatterjee, and R. Brayton, "Improvements to technology mapping for LUT-based FPGAs", *IEEE Transactions on CAD*, 2007. ([PDF](#))
- K. Keutzer, "DAGON: Technology Binding and Local Optimization by DAG Matching," *ACM/IEEE Design Automation Conference*, 1987, pp. 341-347. ([PDF](#))
- J.H. Anderson and F. N. Najm, "Power-Aware Technology Mapping for LUT-Based FPGAs," *IEEE International Conference on Field-Programmable Technology (FPT)*, Hong Kong, 2002, pp. 211 - 218. ([PDF](#))
- J. Anderson, Q. Wang, C. Ravishankar, "Raising FPGA logic density through synthesis-inspired architecture," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, Vol. 20, No. 3, pp. 537-550, March 2012. ([PDF](#))
- Jin Hee Kim, Jason Helge Anderson: Post-LUT-Mapping Implementation of General Logic on Carry Chains Via a MIG-Based Circuit Representation. *FPL 2021*: 334-340 ([PDF](#))

- Lecture 11: Technology Mapping (Dynamic Programming) [continued], Logic Synthesis, Linear Programming

- Background Literature:

- A. Mishchenko, S. Chatterjee, and R. Brayton, "DAG-Aware AIG Rewriting", *DAC*, 2006. ([PDF](#))
- Wei, Bhilare, Waqar, Anderson, "CAD Techniques for NoC-Connected Multi-CGRA Systems," *HEART*, 2024. ([PDF](#))

- Lecture 12: High-Level Synthesis (Linear Programming)

- Philippe Coussy, Daniel D. Gajski, Michael Meredith, Andr?s Takach: An Introduction to High-Level Synthesis. *IEEE Design & Test of Computers* 26(4): 8-17 (2009) ([PDF](#))
- R. Nane, V.-M. Sima, F. Ferrandi, C. Pilato, J. Choi , B. Fort, A. Canis, Y.T. Chen, H. Hsiao, S. Brown, J.H. Anderson, Koen Bertels, "A survey and evaluation of FPGA high-level synthesis tools," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, October 2016. ([PDF](#))
- Jason Cong, Zhiru Zhang: An efficient and versatile scheduling algorithm based on SDC formulation. *DAC* 2006: 433-438 ([PDF](#))
- Andrew Canis, Jongsok Choi, Mark Aldham, Victor Zhang, Ahmed Kammoona, Tomasz S. Czajkowski, Stephen Dean Brown, Jason Helge Anderson: LegUp: An open-source high-level synthesis tool for FPGA-based processor/accelerator systems. *ACM Trans. Embedded Comput. Syst.* 13(2): 24 (2013) ([PDF](#))
- Jae-Gon Kim; Yeong-Dae Kim, "A linear programming-based algorithm for floorplanning in VLSI design," *Computer-Aided Design of Integrated Circuits and Systems*, *IEEE Transactions on* , vol.22, no.5, pp.584,592, May 2003 ([PDF](#))